MILITARY SPECIFICATION

DETONATOR, ELECTRIC, T24E1
LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of electric detonator designated as Detonator, Electric, T24E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-L-201 - Lead, Sheet.
UU-B-536 - Board, Binders.
QQ-S-00640 - Steel, Carbon; Sheet and Strip.

MILITARY

MIL-P-116 - Packaging and Packing for Overseas Shipment.
MIL-B-117 - Bags, Interior Packaging.
MIL-A-148 - Aluminum Foil.
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-14077 - Polytetrafluoroethylene (Teflon), Molding and Extruding Material.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-317 - Detonator Output Measurement by the Lead Disc Test.

FSC 1345

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DRAWINGS

ORDNANCE CORPS

P-86242 - Detonator, Electric, T24E1. Assembly and Details.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-8781490 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78 - Interstate Commerce Commission

Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1940 Edition - latest revision) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publications should cite "49 CFR 71-78 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonators.-The detonators shall comply with all requirements specified on Drawing (dwg.) P-86242, and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide.-The moisture content of the lead azide, at the loading station at the time of loading the detonator, shall not exceed 0.50 percent when tested as specified in 4.3.1.

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3.4 Moisture content of PETN.-The moisture content of the PETN at the loading station at the time of loading the detonator, shall not exceed 0.50 percent when tested as specified in 4.3.2.

3.5 Resistance between shorted lead wires and cup.-The detonators shall have an electrical resistance of not less than 50 megohms when tested as specified in 4.3.4.

3.6 Functioning.

3.6.1 Time.-The detonators shall fire on the first attempt within 120 microseconds when tested as specified in 4.3.5.1.

3.6.2 Output.-The detonators shall fire on the first attempt and shall cause perforation of the lead disc of not less than 3/16 inch in diameter when subjected to the lead disc test in accordance with Standard MIL-STD-317 when tested as specified in 4.3.5.2.

3.6.3 Duds.-The detonators shall fire on the first attempt when tested as specified in 4.3.5.3.

3.7 Pre-pulse.-Detonators to be used in the T37E9 Mine shall not function when subjected to an electrical level of 64 ergs or less when tested as specified in 4.3.5.4.

3.8 Workmanship-All parts shall be free from chips, dirt, grease, rust and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring, and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2.2 and 4.3, and the other

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documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings, and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of

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contractor-accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. Specification MIL-G-2550 shall apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed-upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of detonators loaded and assembled by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Parts of one interfix lot number from one manufacturer.
b. Lead azide from not more than one lot.
c. PETN from not more than one lot.
d. Lead styphnate from not more than one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Bridge and plug assembly, prior to application of spot charge (see dwg. P-86242).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Length of wires to bridge side of plug	Gage
102.	Bridge wire burned or broken	Visual
103.	Excess solder, flux or weld flash	Visual

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Minor: None defined.

4.2.2.2 Bridge and plug assembly (see dwg. P-86242)

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.65 percent	
101.	Spot charge on plug	Visual
102.	Cracks or voids in spot charge	Visual
103.	Bridge wire not totally covered by spot charge	Visual
104.	Evidence of poor workmanship (see 3.8)	Visual
105.	Plug cracked or broken	Visual
Minor: None defined.		

4.2.2.3 Assembly, prior to inserting priming charge (see dwg. P-86242).

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.25 percent	
101.	Distance to top of base charge	Gage
102.	Evidence of poor workmanship (see 3.8)	Visual
Minor: None defined.		

4.2.2.4 Assembly, prior to inserting bridge and plug assembly (see dwg. P-86242).

Categories	Defects	Method of inspection
Critical: None defined.		
Major:	AQL 0.25 percent	
101.	Priming charge missing	Visual
102.	Ferrule missing	Visual
Minor: None defined.		

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4.2.2.5 Assembly (see dwg. P-86242).

Categories	Defects	Method of inspection
Critical:		
	1. Lead wires improperly twisted	Visual
	2. Insulation not removed from lead wires	Visual
	3. Lead wire broken (when applicable)	Visual
	4. Cup cracked or split	Visual
Major:	AQL 0.65 percent	
	101. Resistance of lead circuit	Test (see 4.3.3)
	102. Resistance between shorted lead wires and cup (see 3.5)	Test (see 4.3.4)
	103. Length of loaded cup, maximum (max.)	Gage
	104. Outside diameter of cup, max	Gage
	105. Lead wire not tinned	Visual
	106. Crimp not full 360 degrees	Visual
	107. Varnish seal missing on crimp joint	Visual
Minor:	AQL 1.5 percent	
	201. Evidence of poor workmanship (see 3.8)	Visual
	202. Deep scratch or groove on cup	Visual
	203. Marking misleading or unidentifiable	Visual

4.2.2.6 Carton, sealed (see section 5).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
	101. Carton damaged so that contents are exposed or liable to become exposed	Visual
Minor:	AQL 0.40 percent	
	201. Marking misleading or unidentifiable	Visual

4.2.2.7 Bag, barrier, sealed (see section 5).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
	101. Waterproofness destroyed by rip, tear, puncture, or separation of heat seal	Visual
Minor:	AQL 0.40 percent	
	201. Marking misleading or unidentifiable	Visual

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4.2.3 Testing.

4.2.3.1 Pre-pulse (see 3.7) Critical defect.-This test shall be performed 100 percent. Any detonator assembly which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.5.4 using equipment in accordance with 4.2.4. This test shall follow all other tests.

4.2.3.2 Functioning.-One hundred and eighty-five (185) detonators shall be selected for the following tests.

4.2.3.2.1 Time (see 3.6.1) Major defect.-Thirty-five (35) detonators shall be subjected to this test. The test shall be performed as specified in 4.3.5.1 using equipment in accordance with 4.2.4. If the mean time ($\bar{X}$) plus the product of 2.27 and the standard deviation (s) exceeds 120 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.2, or equivalent. Duds shall not enter into the computation for standard deviation. Duds shall be replaced by detonators from the 185 selected in accordance with 4.2.3.2.

4.2.3.2.2 Output (see 3.6.2) Major defect.-Fifty (50) detonators shall be subjected to this test. The test shall be performed as specified in 4.3.5.2 using equipment in accordance with 4.2.4. If two or more detonators fail to adequately perforate the lead disc, the lot shall be rejected. If one detonator fails to adequately perforate the lead disc, a retest shall be permitted using the balance of the detonators. If, during the retest, one or more detonators fail to adequately perforate the lead disc, the retest may be stopped and the lot shall be rejected. During the retest the Government inspector shall keep a record of any duds that may occur. Duds shall be replaced by detonators from the 185 selected in accordance with 4.2.3.2.

4.2.3 2.3 Duds (see 3.6.3) Major defect.-This test shall be performed as specified in 4.3.5.3 using equipment in accordance with 4.2.4. If a retest is not required in the output phase, the balance of the 185 detonators shall be subjected to the test for duds. If a retest is conducted in the output phase, the acceptance, retest, or rejection for duds shall be based on the results of the combined time and output phase tested. If two duds are found in the detonators tested in 4.3.5.1, 4.3.5.2, and 4.3.5.3, a retest for the duds shall be permitted using 370 detonators. The lot shall be rejected anytime during the retest when one detonator fails to function on the first attempt. If three duds are observed during the testing of the 185 detonators, the lot shall be rejected without retest.

4.2.3.3 Moisture content of lead azide (see 3.3) Major defect.-A two-gram specimen shall be selected from each container of lead azide at the loading station at the time of loading and subjected to the test specified in 4.3.1. If any specimen fails to comply with the applicable requirement, the container from which it was taken, shall not be used in the loading.

4.2.3.4 Moisture content of PETN (see 3.4) Major defect.-A five-gram specimen shall be taken from each container of PETN at the loading station at the time of loading and subjected to the test specified in 4.3.2. If any specimen fails to meet the applicable requirement, the container from which it was taken, shall not be used in the loading.

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4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.3, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment identified by Index of Inspection Equipment Lists No. 8781490, or approved equivalents thereto. When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring, and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration shall be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Moisture content of lead azide.-A tared, dried, aluminum dish, two inches in diameter, shall be covered with a sheet conforming to Specification MIL-P-14077, (see 6.5), 2 1/2 inches square, and sufficiently flat to form a tight cover. A sample of approximately two grams shall be transferred to the moisture dish while both the cover and the dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95 degrees plus or minus ($\pm$) 2 degrees Centigrade ($^{o}C.$) for two hours. The cover shall be placed in the sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The cover shall be replaced on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{(A-B)\ 100}{A}$$

where:

A - the weight of the sample before drying.
B - the weight of the sample after drying.

4.3.2 Moisture content of PETN.-Approximately five grams of PETN shall be placed in a tared, glass-stoppered weighing dish. The dish and its contents shall then be weighed, dried at 100 $\pm$ 2$^{o}C.$ for one hour, cooled in a desiccator and weighed. The loss in weight shall be calculated to the percentage of moisture in the PETN.

4.3.3 Resistance of lead circuit.-Before shorting the lead wires, the resistance of the detonators shall be measured with a specified ohmmeter having an output of not more than 250 micro-amperes, an accuracy not less than five percent of full scale value, and using not more than 1 milliampere of current. The electrical resistance shall be measured to determine compliance with the requirement specified on the applicable drawing.

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4.3.4 Resistance between shorted lead wires and cup.-A potential of 500 volts, root mean square (r.m.s.), 60 cycles per second, shall be applied for two seconds, max. between the shorted lead wires and the cup. The electrical resistance shall be measured to determine compliance with the requirement specified in 3.5.

4.3.5 Functioning.

4.3.5.1 Time.-The detonators shall be temperature-conditioned to a temperature of 70 degrees Fahrenheit (^{o}F.) $\pm$ 10°F., then assembled to the electric detonator test set specified in Standard MIL-STD-317 and tested at that temperature by applying the energy from a 16 microfarad capacitor charged to 2.5 volts, max. Functioning time and any duds that may occur shall be recorded. (The functioning time is the elapsed time from the instant the current is applied until the explosion occurs). Duds shall be replaced by detonators from the 185 selected.

4.3.5.2 Output.-The detonator shall be assembled in the above test apparatus and tested at a temperature of 70°F. by applying an energy of 500 ergs, max., using a capacitor charged to 2.5 volts, max. Observation shall be made to determine compliance with 3.6.2. Duds, if any, shall be recorded. Duds shall be replaced by detonators from the 185 selected.

4.3.5.3 Duds.-The test shall be performed by applying the energy of 500 ergs from a capacitor charged to 2.5 volts, max. Duds, if any, shall be recorded.

4.3.5.4 Pre-pulse.-The detonators shall be pre-pulsed with 64 ergs of electrical energy by directly discharging a five microfarad capacitor, with a tolerance of $\pm$ 5 percent, which has been charged to 1.60 volts, through the detonator. A record of this test shall be annotated on the data cards.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level B.

5.1.1.1 Rosette wrap.-A bundle of 25 detonators shall be made, lead wires out straight, using the lead wires from one detonator to secure the bundle. Each bundle of detonators shall be wrapped in a sheet of approximately eight by eight by .0015 inch-thick aluminum foil complying with Specification MIL-A-148, so that all detonators and lead wires in a bundle are completely encased by the foil. Five bundles of detonators, totaling 125 detonators, shall be packed in a folding carton, set-up box, or comparable container, made of material with a minimum board thickness of 0.03 inch. When destined for shipment and storage, uncovered, in excess of 180 days from the loading date shown on the packing box to the date assembled into end items, the following shall be included: Each carton, set-up box, or comparable container, after blunting corners, shall be placed in a water-vapor-proof bag manufactured of barrier material in accordance with Type I, Class 2, Specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and sealed in accordance with manufacturer's instructions with seal located at edge of bag to provide for a minimum of two subsequent seals, if required. The seals shall comply with Specification MIL-B-117. Alternatively, Method 1A2, in accordance with Specification MIL-P-116, may be used with package overwrapped with 30 pound Kraft commercial paper.

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5.1.1.2 Pocket assembly, alternate pack.-The detonators shall be packed in pocket assemblies manufactured in accordance with figure 1 (as applicable) made of .0015 inch aluminum foil complying with Specification MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five pockets. The adhesive used for the sealing of the two faces of aluminum foil shall be suitable for the application. The strength of the seal shall be sufficient to allow the insertion of the dowel (see figure 1) without separation of the seal. The two faces of any seal shall not be misaligned by more than 1/16 inch when the pouch is formed. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end similar to that shown on figure 1, (as applicable). Each pocket may, however, be opened by other suitable methods which will result in providing a satisfactory pocket for insertion of the detonator without injury to the material. One T24E1 Electric Detonator shall be inserted, detonator end down, in each pocket. The lip of the pocket assembly shall be folded over to form a closed pouch. Twenty-five pocket assemblies, or a total of 125 detonators, shall be packed in a folding carton, set-up box, or comparable container made of material with a minimum board thickness of .03 inch. When destined for shipment and storage uncovered in excess of 180 days from the loading date shown on the packing box to the date assembled into end items, the following shall be included: Each carton, set-up box, or comparable container, after blunting corners, shall be placed in a water-vapor-proof bag manufactured of barrier material in accordance with Type I, Class 2, Specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and sealed in accordance with manufacturer's instructions with seal located at edge of bag to provide for minimum of two subsequent seals, if required. The seals shall comply with Specification MIL-B-117. Alternatively, Method 1A2 in accordance with Specification MIL-P-116, may be used with package overwrapped with 30 pound Kraft paper, commercial.

5.1.2 Level C.-Unless otherwise specified, T24E1 detonators shall be preserved and packaged as specified in 5.1.1, except that no water-vapor-proof protection is required.

5.2 Packing

5.2.1 Level B.-A suitable number of containers, each containing 125 detonator assemblies, shall be packed in an outer case in accordance with Tariff 10, Interstate Commerce Commission Regulations for transportation of Explosives and Other Dangerous Articles, paragraph 73.66. (See Code of Federal Regulations 49 CFR 71-78).

5.2.2 Level C.-Unless otherwise specified, T24E1 detonators shall be packed as specified in 5.2.1.

5.3 Marking.-Interior packages and exterior containers shall be marked in accordance with Standard MIL-STD-129. When the barrier bag is used, the marking shall be applied directly to the bag. Cartons may be marked with labels or printed directly on the carton. When alternative method of water-vapor protection is used, three labels shall be required, one applied to the carton, and one each applied to inner and outer wrappers. The ICC Designation shall be (see note): Electric Blasting Caps - Handle Carefully.

NOTE: "The number of caps packed in one outer pack shall be shown in the marking."

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Each inner carton, set-up box, or comparable container, shall be marked: "PATENTED - U.S. Patent 2,389,086." (see 6.4).

5.4 Data cards.-Data card information shall be as specified in Specification MIL-G-2550 and 4.3.5.4.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Levels of protection.
c. Unit, intermediate and exterior package quantities, when applicable.

6.2 The following formula, or its equivalent, shall be used to compute the standard deviation:

$$S = \sqrt{\frac{\Sigma(X-\bar{X})^2}{n-1}}$$

where:

S - standard deviation.
Σ - sum of.
X - individual observed time.
$\bar{X}$ - sample mean time.
n - sample size.

6.3 Check test for possible deterioration.-This test is designed to prevent the loading into a major ammunition item of detonators which have deteriorated. The test should be performed if the detonators have been stored under normal condition for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonator is loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract. Two hundred (200) detonators shall be selected and tested as follows:

6.3.1 Function time and duds.

6.3.1.1 Procedure.-Forty (40) detonators shall be subjected to the test specified in 4.3.5.1. Observation shall be made for the functioning time and all duds that may occur will be recorded. Duds will be replaced by detonators from the 200 selected.

6.3.1.2 Rejection.-If the mean time ($\bar{X}$) plus 2.03 times the standard deviation (s) exceeds 120 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in paragraph 6.2, or equivalent. Duds shall not enter into the calculation for standard deviation.

6.3.2 Output and duds.

6.3.2.1 Procedure.-Fifty (50) detonators shall be subjected to the test specified in 4.3.5.2. Observation shall be made for the output and the number of duds that may occur will be recorded. Duds will be replaced by detonators from the 200 selected.

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6.3.2.2 Retest and rejection.-If two detonators fail to adequately perforate the lead disc, a retest shall be permitted using the balance of the detonators selected in 6.3. If, during the retest, one more detonator fails to adequately perforate the lead disc, the retest shall be stopped and the lot rejected. During the retest, the number of duds that may occur shall be recorded. If three or more detonators fail to adequately perforate the lead disc during the original test, the lot shall be rejected without retest.

6.3.3 Duds.

6.3.3.1 Procedure.-If no retest is required in the output phase, the balance of the detonators selected in 6.3 shall be subjected to the test for duds as specified in 4.3.5.3.

6.3.3.2 Retest and rejection.-If more than two, but not more than six duds are found, a retest shall be permitted using double the number of detonators selected in 6.3. The lot shall be rejected during retest as soon as the total number of duds observed in 6.3.1, 6.3.2 and 6.3.3 is equal to seven. If in the original sample of 200, seven or more duds are observed, the lot shall be rejected without retest.

6.4 The United States Government has negotiated a license agreement covering manufacture of items within the scope of this patent with Hercules Powder Company.

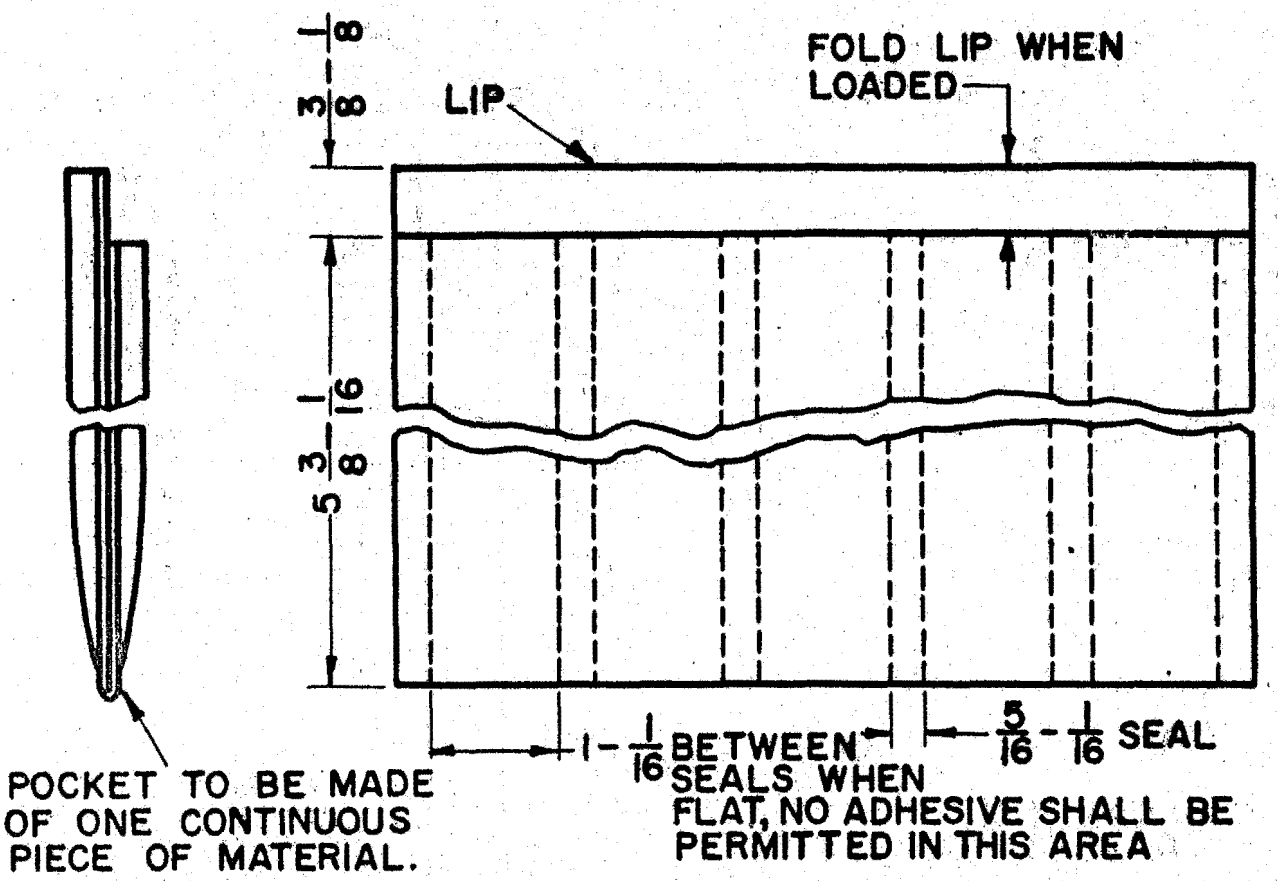
6.5 Polytetrafluoroethylene.-"Teflon" - a product of E.I.DuPont DeNemours and Company, Wilmington, Delaware, has been found satisfactory.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

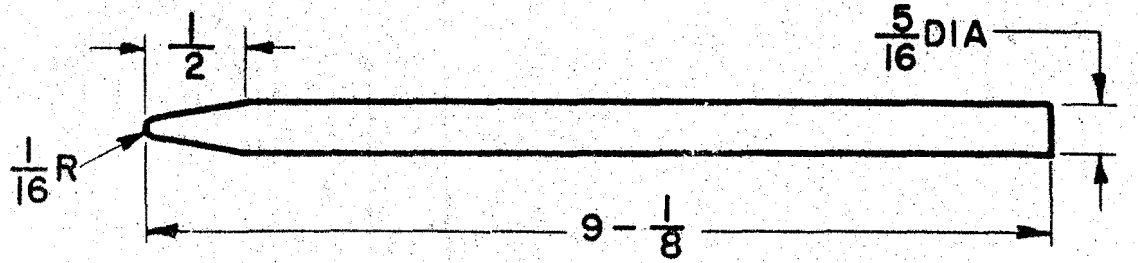
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POCKET ASSEMBLY



DOWEL SHALL BE SMOOTH WITH NO SHARP CORNERS

DOWEL

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FIGURE 1